Feedback Linearization Control for a 3-DOF Flexible Joint Elbow Manipulator

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***Abstract*-This paper is concerned with the feedback linearization control of a flexible joint anthropoid robot. The theory to derive a linearizing control law for a certain type of dynamic equations is reviewed and the model for a particular manipulator is derived. Based on this model a linearization controller is designed and its tracking performance is compared with classical controllers.**

I. INTRODUCTION

Feedback linearization control of manipulators has attracted the attention of many authors [1]-[2]. The necessary and sufficient conditions for a single input system to be feedback linearizable are explored in [3]. Reference [1] deals with a single link flexible joint manipulator dynamical model and shows that this model satisfies the conditions to be feedback linearizable. The paper derives a feedback linearizing control law for this single input system. Reference [2] treats the n-link case for a certain type of dynamic model and uses the structure of the controller derived for the single link flexible manipulator to devise a feedback linearizing control for this model. These derivations are reviewed in Sec. 2. The derivation of the dynamical model of a 3-DOF elbow manipulator is presented in Sec. 3. Section 4 includes the simulation results for this manipulator and compares tracking performances of PD control, PID control, PD control with gravity compensation, inverse dynamics control to the performance of the feedback linearization control. Simulations are run with and without joint flexibility in order to observe the worsening effect of elasticity. Conclusions are given in Sec.5.

II. FEEDBACK LINEARIZATION

The idea of feedback linearization is to use a nonlinear control law such that the nonlinear plant and the nonlinear control together behave linear and controllable in the closed loop (Fig.1). Designing a linearizing control law is referred to as the inner loop control. Controlling the linear plant established by the inner loop control is called the outer loop control.

The three operations; coordinate changes in the state space, coordinate changes in the control space, and feedback are the basis of the so called feedback equivalence. Broadly, a system is called feedback equivalent to some other system if state space equations of this system can be modified, by the use of appropriate operations named above, to the state space equations of the second system. It can be shown that feedback equivalence really defines equivalence classes. For single input linear systems, an equivalence relation can be defined as follows.

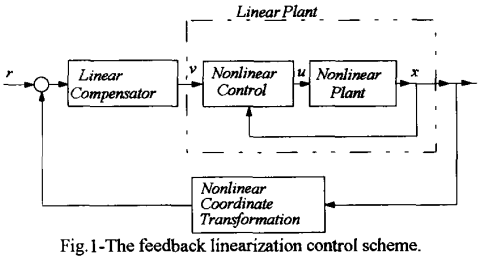


Fig.1-The feedback linearization control scheme.

If S_1 and S_2 are two systems with

$$\boldsymbol{x}' = \boldsymbol{f}(\boldsymbol{x},u),\ \boldsymbol{y}' = \boldsymbol{g}(\boldsymbol{y},v), \tag{1}$$

respectively, S_1 is said to be related to S_2 if there exists a C^∞ diffeomorphism

$$\boldsymbol{T}: R^{n+1} \rightarrow R^{n+1} \tag{2}$$

such that

$$T_1(\boldsymbol{x},u) = y_1,\ T_2(\boldsymbol{x},u) = y_2,\ \ldots,\ T_n(\boldsymbol{x},u) = y_n, T_{n+1}(\boldsymbol{x},u) = v \tag{3}$$

with $\boldsymbol{y}$ and v satisfying the equations defining S_2 , that is,

$$\boldsymbol{y}' = \boldsymbol{g}(\boldsymbol{y},v). \tag{4}$$

It is well known that [4] every linear, time-invariant, controllable single input system is feedback equivalent to the system represented by the state space equation

$$\begin{pmatrix} \dot{y}_1 \\ \dot{y}_2 \\ \vdots \\ \dot{y}_n \end{pmatrix} = \begin{pmatrix} y_2 \\ \vdots \\ y_n \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} v. \tag{5}$$

It is therefore to be expected that every feedback linearizable single input system is transformable to the system above.

Using results from differential geometry, [1] shows that feedback linearization can be achieved for the single link manipulator with the dynamical equations

$$I q_1'' + Mgl \sin(q_1) + k(q_1 - q_2) = 0,$$
$$J q_2'' - k(q_1 - q_2) = u \tag{6}$$

where I is the inertia of the link about the axis of rotation, M is the mass of the link, l is the distance between the joint and the center of the mass of the link, g is the gravity constant, J is the motor inertia and k is the stiffness constant representing the joint flexibility. u is the generalized torque input, q_1 is the link angular position and q_2 is the angle of the shaft before the joint flexibility. Unlike for the rigid single link manipulator, the control u cannot be chosen to cancel the nonlinearity [5] since it appears in the first equation of (6). Feedback linearization control provides us with a tool to deal with this problem.

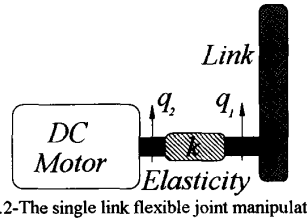


Fig.2-The single link flexible joint manipulator.

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By assigning the state variables as

$x_1 \equiv q_1,\ x_2 \equiv q_1',\ x_3 \equiv q_2,\ x_4 \equiv q_2'$

and defining $\boldsymbol{x} \equiv (x_1\, x_2\, x_3\, x_4)^T$, we can represent the dynamics of the single link flexible manipulator in state space form as

$$\boldsymbol{x'} = \boldsymbol{f(x)} + \boldsymbol{g(x)}\, u \tag{7}$$

with

$$\boldsymbol{f(x)} = \begin{pmatrix} x_2 \\ -(Mgl/I)\sin(x_1) - (k/I)(x_1 - x_3) \\ x_4 \\ (k/J)(x_1 - x_3) \end{pmatrix},$$

$$\boldsymbol{g(x)} = (0\ 0\ 0\ 1/J)^T. \tag{8}$$

In [1] it is shown that $\boldsymbol{f(x)}$ and $\boldsymbol{g(x)}$ for this manipulator satisfy the conditions in [3] for the dynamics to be feedback linearizable. The dynamics of this manipulator is therefore feedback equivalent to the system in (5). Furthermore, it is shown that the transformation

$$y_1 = T_1 = x_1,\ y_2 = T_2 = x_2,$$

$$y_3 = T_3 = y_2' = -\frac{Mgl}{I}\sin(x_1) - \frac{k}{I}(x_1 - x_3), \tag{9}$$

$$y_4 = T_4 = y_3' = -\frac{Mgl}{I}\cos(x_1)x_2 - \frac{k}{I}(x_2 - x_4),$$

$$v = \langle dT_4, \boldsymbol{f} \rangle + \langle dT_4, \boldsymbol{g} \rangle\, u,$$

which defines a diffeomorphism, results in a linear controllable system

$$y' = \boldsymbol{A}y + \boldsymbol{b}v \tag{10}$$

with

$$y = (y_1\, y_2\, y_3\, y_4)^T,$$

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix},\quad \boldsymbol{b} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{11}$$

In the expression above $\langle ... \rangle$ denotes the inner product and dT_4 stands for the gradient of T_4.

The feedback linearizing control is then

$$u = \frac{IJ}{k}(v - a(x)) \tag{12}$$

with

$$a(x) = \frac{MgL}{I}\sin(x_1)(x_2^2 + \frac{MgL}{I}\cos(x_1) + \frac{k}{I}) + \frac{k}{I}(x_1 - x_3)(\frac{k}{I} + \frac{k}{J} + \frac{MgL}{I}\cos(x_1)). \tag{13}$$

The new state variables y_1, y_2, y_3 and y_4 are the link position, velocity, acceleration and jerk respectively. If we are given a desired trajectory y_1^d, we can choose an outer loop control

$$v = y_4^d - C_1(y_1 - y_1^d) - C_2(y_2 - y_2^d) - C_3(y_3 - y_3^d) - C_4(y_4 - y_4^d) \tag{14}$$

where y_{i+1}^d is the i-the derivative of the desired link position.

This control, with $e = y_1 - y_1^d$, results in the error dynamics

$$\frac{d^4e}{dt^4} + C_4\frac{d^3e}{dt^3} + C_3\frac{d^2e}{dt^2} + C_2\frac{de}{dt} + C_1 e = 0 \tag{15}$$

By choosing the gains C_i such that the error characteristic equation has all its roots in the left half plane, we can assure asymptotic tracking. Putting two of the poles far in the left half plane and the remaining two in such a way that they coincide on the negative real axis, a controller for a critically damped step response can be designed.

The multi-input case for feedback linearization is complicated to analyze [2], [6]. In the case of flexible joint robotic manipulators with DC motors placed at the joints and all joints being revolute, we can follow the guidelines obtained in the single link example and avoid more complicated analysis.

For an n-link rigid manipulator of the same configuration, defining θ_{m_i} as the angle and r_i as the gear reduction ratio of motor i, we can choose the generalized coordinates q_i as $q_i = \theta_{m_i} r_i$. Defining further J_{m_i} as the inertia, B_{m_i} as the friction, K_{m_i} as the torque, K_{b_i} as the back emf constants, R_i as the armature resistance, v_i as the motor voltage for motor i and ignoring the electrical; time constant of each motor, we can model the dynamics of the manipulator as

$$(\boldsymbol{D(q)} + \boldsymbol{J})\boldsymbol{q''} + \boldsymbol{C(q,q')q'} + \boldsymbol{Bq'} + \boldsymbol{g(q)} = \boldsymbol{u}. \tag{16}$$

In this equation $\boldsymbol{q}$ is the vector of generalized coordinates, $\boldsymbol{D(q)}$ is the inertia matrix of the manipulator, $\boldsymbol{J}$ and $\boldsymbol{B}$ are diagonal matrices with elements (J_{m_i}/r_i^2) and $(B_{m_i} + K_{b_i}K_{m_i}/R_i)/r_i^2$ respectively, $\boldsymbol{C(q,q')q'}$ represents the Coriolis and centrifugal terms, $\boldsymbol{g(q)}$ accounts for the gravity terms and $\boldsymbol{u}$ is the vector of generalized forces constructed by the components $(K_{m_i}/r_i R_i)\, v_i$. The expression results from the Euler-Lagrange equations of the manipulator together with the actuator dynamics equations.

Having this model for the rigid manipulator ,it can be shown [5] that the dynamics equations of the flexible joint robot with the same configuration and with the joint stiffness constant k_i for joint i are

$$\boldsymbol{D(q_1)q_1''} + \boldsymbol{C(q_1,q_1')q_1'} + \boldsymbol{g(q_1)} + \boldsymbol{K(q_1 - q_2)} = \boldsymbol{0},$$
$$\boldsymbol{J q_2''} + \boldsymbol{Bq_2'} - \boldsymbol{K(q_1 - q_2)} = \boldsymbol{u}. \tag{17}$$

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In this expression, q_1 is the vector $(q_1,q_3,...,q_{2n-1})^T$ of link positions, q_2 is $(q_2,q_4,...,q_{2n})^T$ with q_{2i-1} ≡angle of link i and $q_{2i} \equiv \theta_{m_i} r_i$. $K \equiv diag(k_1,...,k_n)$. Ignoring damping and defining $h \equiv C q_1' + g$, we can obtain the dynamics equations of the form

$$D(q_1) + h(q_1,q_1') + K(q_1 - q_2) = 0,$$
$$J q_2'' - K(q_1 - q_2) = u. \quad (18)$$

As in the single link example, the nonlinearity enters the first equation and it is impossible to use the so called inverse dynamics control [5] to cancel it.

It is shown in [2] that the dynamic model (18) is feedback linearizable. By defining $x_1 = q_1$, $x_2 = q_1'$, $x_3 = q_2$, $x_4 = q_2'$, we can obtain the state space equations

$$x_1' = x_2, \quad x_2' = -D(x_1)^{-1}(h(x_1,x_2) + K(x_1 - x_3)), \quad (19)$$
$$x_3' = x_4, \quad x_4' = J^{-1}K(x_1 - x_3) + J^{-1}u.$$

Using the multi-variable analog of the transformation (9) of the single input case, a feedback linearization control law can be obtained.

$$y_1 = T_1(x) = x_1, \qquad y_2 = T_2(x) = x_2,$$
$$y_3 = T_3(x) = x_2' = -D(x_1)^{-1}(h(x_1,x_2) + K(x_1 - x_3)), \quad (20)$$

$$y_4 = T_4(x) = y_3' = -\frac{d}{dt}(D(x_1)^{-1})(h(x_1,x_2) + K(x_1 - x_3))$$
$$- D(x_1)^{-1}\{\frac{\partial h}{\partial x_1}x_2 + \frac{\partial h}{\partial x_2}[-D(x_1)^{-1}(h(x_1,x_2) + K(x_1 - x_3))]$$
$$+ K(x_2 - x_4)\}$$
$$\equiv a_4(x_1,x_2,x_3) + D(x_1)^{-1}K x_4. \quad (21)$$

The outer loop control v and the fourth derivative of the link position vector satisfy $v = y_4'$. This results in an expression

$$v = \frac{\partial a_4}{\partial x_1}x_2 - \frac{\partial a_4}{\partial x_2}D^{-1}[h + K(x_1 - x_3)] + \frac{\partial a_4}{\partial x_3}x_4$$
$$+ \frac{d}{dt}(D^{-1})K x_4 + D^{-1}K[J^{-1}K(x_1 - x_3) + J^{-1}u] \quad (22)$$
$$\equiv a(x) + b(x)u. \quad (23)$$

So, the linearizing control is

$$u = b(x)^{-1}[v - a(x)]. \quad (24)$$

The resulting system is then

$$\dot{y} = \begin{pmatrix} 0 & I & 0 & 0 \\ 0 & 0 & I & 0 \\ 0 & 0 & 0 & I \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ I \end{pmatrix} v. \quad (25)$$

The equation above is completely decoupled and we can specify the poles of each decoupled SISO system by using state feedback. As in the single link case, the new state vectors represent the position, velocity, acceleration and jerk of the manipulator links. We can use diagonal gain matrices and the control

$$v = y_4^d - C_1(y_1 - y_1^d) - C_2(y_2 - y_2^d) - C_3(y_3 - y_3^d) - C_4(y_4 - y_4^d) \quad (26)$$

with the obvious definitions for y_i^d, in the outer loop, to assure asymptotic tracking of reference trajectories.

III. THE DYNAMICAL MODEL FOR THE 3-DOF ELBOW MANIPULATOR

In this section, a dynamical model of the form (18) will be derived for a 3-DOF robot with DC motors placed at the three joints. The derivation is based on the discussion in [5].

The following terminology is used.

l_i≡the length of link i, m_i≡the mass of link i,

l_i'≡the distance from joint i to the center of mass of link i,

I_i≡the inertia matrix for link i computed about its center of mass,

$A_i \equiv (I)_{11}, E_i \equiv (I)_{22}, I_i \equiv (I)_{33}$.

Since the motors are located at the joints, the joint variables (joint angles) can be used as generalized coordinates. So, the inertia matrix D can be expressed [5] as

$$\sum_{i=1}^{n}(m_i J_{v_{c_i}}(q)^T J_{v_{c_i}}(q) + J_{\omega_i}(q)^T R_0^i(q)\, I_i\, R_0^i(q)^T J_{\omega_i}(q)) \quad (27)$$

In the expression above, $J_{v_{c_i}}$ is defined by $v_{c_i} = J_{v_{c_i}} q'$ and J_{ω_i} by $\omega_{c_i} = J_{\omega_i} q'$ where v_{ci} and ω_{c_i} are the translational and angular velocities of the center of mass of link i respectively. R_0^i is the rotation matrix relating the orientation of link i to the base frame. Calculating the individual terms in this expression and putting them together, the inertia matrix with the following entries is obtained.

$$d_{11} = m_2 l_{c2}^2 \cos^2 q_2 + m_3[l_2 \cos q_2 + l_{c3}\cos(q_2 + q_3)]^2$$
$$+ E_1 + A_2 \sin^2 q_2 + E_2 \cos^2 q_2 + A_3 \sin^2(q_2 + q_3) + E_3 \cos^2(q_2 + q_3)$$
$$d_{22} = m_2 l_{c2}^2 \sin^2 q_2 + m_3[l_2^2 + l_{c3}^2 + 2 l_2 l_{c3} \cos q_3] + I_2 + I_3$$
$$d_{23} = d_{32} = m_3[l_{c3}^2 + l_{c3} l_2 \cos q_3] + I_3,$$
$$\text{and } d_{33} = m_3 l_{c3} + I_3 \quad (28)$$

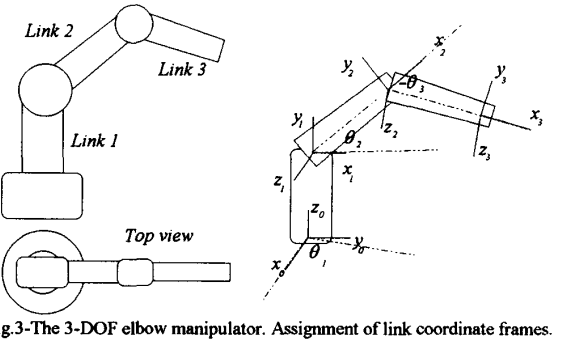


Fig.3-The 3-DOF elbow manipulator. Assignment of link coordinate frames.

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Remaining entries of this matrix are zero.
The nonzero Cristoffel symbols are obtained to be

$$c_{112} = -hc_1,\ c_{113} = -hc_2,\ c_{121} = c_{211} = hc_1,\ c_{131} = c_{311} = hc_2,$$

$$c_{222} = hc_3,\ c_{223} = -hc_4,\ c_{232} = c_{322} = hc_4,\ c_{332} = hc_4 \qquad (29)$$

where

$$hc_1 = \frac{1}{2}\frac{\partial d_{11}}{\partial q_2} = (-m_2 l_{c2}^2 + A_2 - E_2)cosq_2 sinq_2$$

$$+(A_3 - E_3)cos(q_2+q_3)sin(q_2+q_3) \qquad (30)$$

$$+m_3(l_2 cosq_2 + l_{c3}cos(q_2+q_3))(-l_2 sinq_2 - l_{c3}sin(q_2+q_3))$$

$$hc_2 = \frac{1}{2}\frac{\partial d_{11}}{\partial q_3}$$

$$= sin(q_2+q_3)(-m_3 l_{c3} l_2 cosq_2 + (-m_3 l_{c3}^2 + A_3 - E_3)cos(q_2+q_3))$$

$$hc_3 = \frac{1}{2}\frac{\partial d_{22}}{\partial q_2} = m_2 l_{c2}^2 cosq_2 sinq_2 \text{ and } hc_4 = \frac{1}{2}\frac{\partial d_{11}}{\partial q_3} = -m_2 l_2 l_{c3} sinq_3$$

The potential energies of the individual links are

$$V_1 = 0,$$

$$V_2 = m_3 g\, l_{c2} sinq_2,\ V_3 = m_3 g(l_2 sinq_2 + l_{c3} sin(q_2+q_3)). \qquad (31)$$

So, the gravity term in (16) for this manipulator is found to be $\boldsymbol{g} = (\phi_1, \phi_2, \phi_3)^T$ where

$$\phi_1 = 0,\quad \phi_2 = (m_2 l_{c2} + m_3 l_2)g\, cosq_2 + m_3 l_{c3} g\, cos(q_2+q_3) \text{ and}$$

$$\phi_3 = m_3 l_{c3} g\, cos(q_2+q_3). \qquad (32)$$

The components of the vector $\boldsymbol{h}$ in (18) are then

$$h_1 = \phi_1 + 2hc_1 q_1' q_2' + 2hc_2 q_2' q_3'$$

$$h_2 = \phi_2 - hc_1 q_1'^2 + 2hc_4(q_2' q_3' + q_3'^2) + hc_3 q_2'^2$$

$$h_3 = \phi_3 - hc_2 q_1'^2 - hc_4 q_2'^2 \qquad (33)$$

Having calculated $\boldsymbol{D(q)}$ and $\boldsymbol{h(q,q')}$, and by the discussion in Section 2, we know now the dynamical equations of the flexible joint 3-DOF elbow manipulator of the form (18). This model is used in the simulations.

IV. Simulation Results

The control of the 3-DOF elbow manipulator is simulated for rigid and flexible joint cases. In the following, the control methods used will be discussed.

The control law for PD control is

$$\boldsymbol{u = K_p(z^d - z) - K_D z'}. \qquad (34)$$

In this expression, $\boldsymbol{z^d}$ is the desired position. For the rigid case $\boldsymbol{z}$ stands for the vector of joint angles. In the flexible joint case, either the link position vector $\boldsymbol{q_1}$ or the shaft position vector $\boldsymbol{q_2}$ can be used in feedback. The term "shaft position" is used for the angle before the flexibility and after the gear reduction. $\boldsymbol{K_p}$ and $\boldsymbol{K_D}$ are diagonal matrices. Their entries are calculated for each joint independently for critically damped step responses. The system is second order and thus once specified a desired natural frequency ω, with the controller being designed for a critically damped response, the PD gains are uniquely determined. The coupling effects are considered as disturbance. The motor angle is assumed to satisfy

$$J_{eff_i}\Theta_{m_i}'' + (B_{m_i} + K_{b_i}K_{m_i}/R_i)\Theta_{m_i}' = (K_{m_i}/R_i)\,v_i - r_i d_i. \qquad (35)$$

J_{eff_i} is the so called effective inertia constant for link i. This constant is an approximation to the position dependent coefficient $J_{m_i} + r_i^2 d_{ii}(q)$ of θ_{m_i}'' [4]. d_i stands for the disturbance created by the coupling effects. In our case the constants are chosen to be

$$J_{eff_1} = J_{m_1} + r_1^2(m_2 l_{c2}^2 + m_3(l_2 + l_{c3})^2 + E_1 + E_2 + E_3)$$

$$J_{eff_2} = J_{m_2} + r_2^2(m_3(l_2^2 + l_{c3}^2 + 2l_2 l_{c3}) + I_2 + I_3)$$

$$J_{eff_3} = J_{m_3} + r_3^2(m_3 l_{c3} + I_3) \qquad (36)$$

PID control is used similarly, adding an integral term to the control law above.

For the rigid case, for the set point control, with the control law (34) ,in the absence of gravity, it can be shown [5] by Lyapunov analysis that this control law achieves asymptotic tracking. In the presence of gravity terms PD control can not guarantee asymptotic tracing. The modified control law

$$\boldsymbol{u = K_P(q_d - q) - K_D q' + g(q)} \qquad (37)$$

can remove the steady state error. This control cancels gravitational effects. High speed on-line computation capability is required to calculate $\boldsymbol{g(q)}$ and use it in the control. In the absence of gravity, the control scheme above assures asymptotic tracking for the set point control even if there is joint flexibility, with motor variables used in the feedback. But with flexibility, gravity compensation is impossible. Feedback linearization gives us a solution.

The dynamics equations of the rigid manipulator (16) can be written as

$$\boldsymbol{M(q)q'' + h(q,q') = u}. \qquad (38)$$

Although it is in general difficult or impossible to find a control $\boldsymbol{u = f(q,q')}$ which completely linearizes the system in closed loop, in the case of this dynamic model, by choosing

$$\boldsymbol{u = M(q)\,v + h(q,q')} \qquad (39)$$

we can reduce the system equations to $\boldsymbol{q'' = v}$. The control $\boldsymbol{v}$ can be chosen as

$$\boldsymbol{v = -K_0 q - K_1 q + r} \qquad (40)$$

where $\boldsymbol{r}$ is the reference input determined by the given desired trajectory as

$$\boldsymbol{r(t) = q''_d(t) + K_0 q_d(t) + K_1 q'_d(t)}. \qquad (41)$$

This results in the error dynamics

$$\boldsymbol{e''(t) + K_1 e' + K_0 e = 0} \qquad (42)$$

where $\boldsymbol{e = q^d - q}$. This is the inverse dynamics control for robotic manipulators. The choice of the gain matrices as

$$\boldsymbol{K_0 = diag(\omega_1^2, \ldots, \omega_n^2),\ K_1 = diag(2\omega_1, \ldots, 2\omega_n)} \qquad (43)$$

will result in a critically damped system.

The following parameters are used in the simulations.

$l_1 = 0.5$ m, $l_2 = l_3 = 0.4$ m, $l_{c2} = l_{c3} = 0.2$ m, $m_1 = 4$ kg,

$m_2 = m_3 = 3$ kg, $k_i = 30$ Nm/rad, $r_i = 0.2$, $R_i = 1.3\ \Omega$,

$J_{mi} = 0.000039$ Nms^2 /rad $\forall$ i.

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The inertia parameters are computed assuming that the links are cylindrical with radius R (=5cm). Used formulae are.

$E_i = m_i R^2/2,\ A_i = m_i R^2/2,\ E_i = I_i = m_i l_i^2/12$ for i = 2,3.

Simulations are run for two reference trajectories for each controller: One of them is a step reference of 1 rads for each link. For the other reference the desired trajectory is a cosine curve for each position. (References are dashed curves in the figures.) So, the performance of the controllers for time varying reference inputs can be observed.

The rigid manipulator: Links 2 and 3 are under the diverse effect of gravity and the first link is not affected by it. In the set point PD control of the rigid robot, the position of the first link converges to the set value even with low specified natural frequencies whereas there is always a steady state error for link positions 2 and 3. Fig.4 shows the link positions with cosine position references. Being pulled by gravity, the mean values of link 2 and link 3 trajectories are below the mean of their references. PID control reduces the steady state error in these link positions for the set point control. But the cosine tracking performance is not improved significantly by using PID control. Fig.5 compares the set point controls of PD and "PD with gravity compensation" controls. Gravity compensation removes the steady state error for step references but is not successful for time varying trajectories. The exact cancellation of nonlinearities by the inverse dynamics control results in perfect tracking of both step and time varying references (Fig.6).

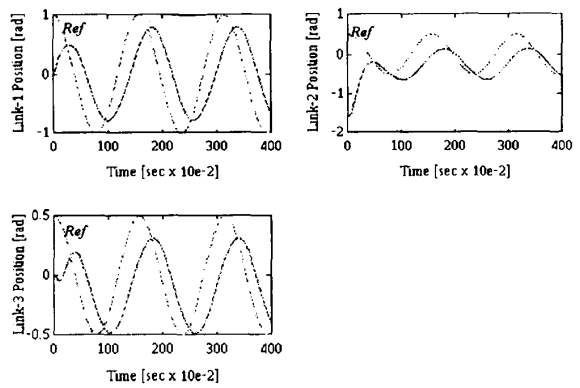


Fig.4-PD control of the rigid manipulator for sinusoidal reference trajectories. ω = 8 rad/sec for each link.

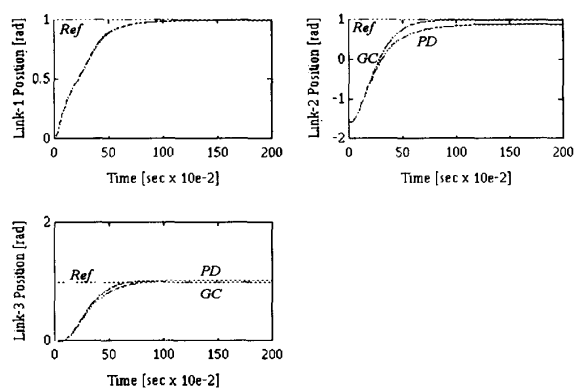


Fig.5-Rigid robot. Comparison of PD control to the PD with gravity compensation scheme. ω = 8 rad/sec for each link .

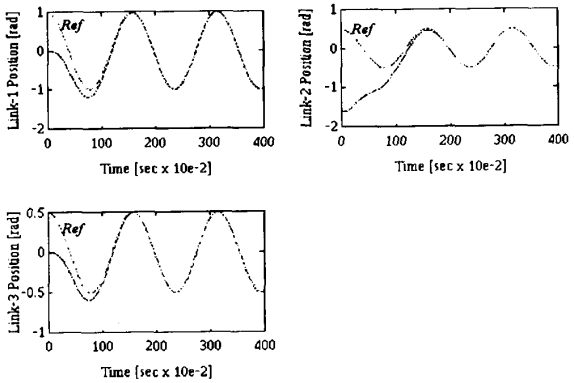


Fig.6- Rigid robot. Inverse dynamics control for sinusoidal link position references. The specified natural frequency is 4 rad/sec for each link.

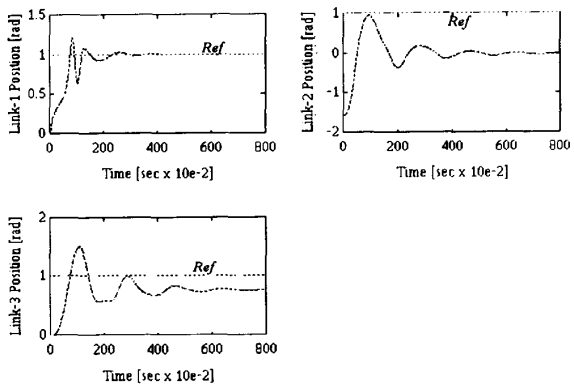


Fig.7- Flexible joint robot. PD control with motor variable feedback. The specified natural frequency is 8 rad/sec for each link.

The flexible joint robot: The PD control with motor variable feedback method results in the link positions shown in Fig.7. For link 1, in the absence of gravity disturbance the position error converges to zero. Big steady state errors observed in links 2 and 3. For the cosine reference, this control cannot be used. As in the rigid case inclusion of integral control and gravity compensation control do not improve the tracking for time varying references. Treating the flexible joint as rigid, the inverse dynamics control can be used with motor variable feedback. Compared to the previous controllers this results in better responses in both step and cosine references (Fig.8). In simulations with higher stiffness constants the position curves get closer to the reference trajectories.

Step response for the feedback linearization control is shown in Fig.9. Critically damped link position curves are obtained. In the sinusoidal reference case (Fig.10,11), shaft position curves are much different than the link position curves and perfect tracking of the reference is observed. This is due to the fact that in the feedback linearization control, joint flexibility is included in the dynamical model and the controlled outputs are the link variables instead of shaft positions. In the previous control methods, the assumption is that joint flexibility is negligible and the link and shaft positions are the same. So the control in the previous methods is done for the shaft variable and from the resulting curves, it can be observed that shaft positions are closer to

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the reference trajectories. In the case of feedback linearization, the shaft before the flexibility adapts its position a way to absorb the oscillations in the fictitious torsional spring representing the flexibility.

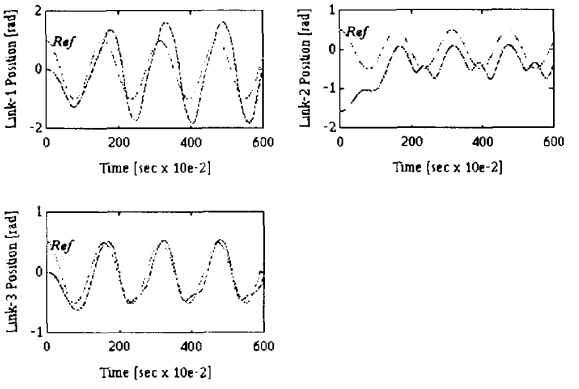


Fig.8- Flexible joint robot. Link positions with inverse dynamics control for sinusoidal position references. $\omega_i = 4\ \forall\ i$.

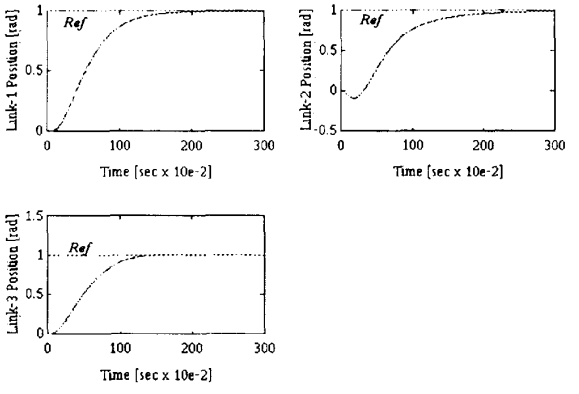


Fig.9-Flexible joint robot. Feedback linearization control for step references. Link positions for system poles at -4,-4,-20,-24.

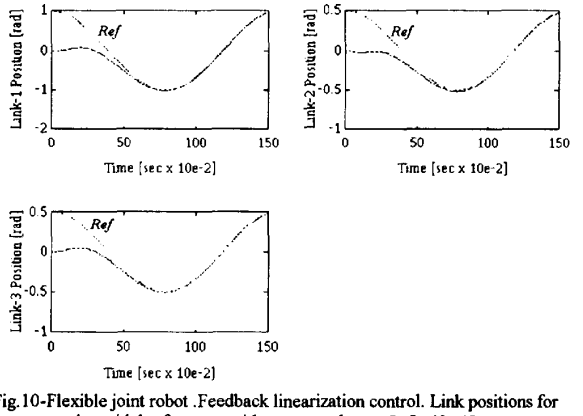


Fig.10-Flexible joint robot .Feedback linearization control. Link positions for sinusoidal references with system poles at -8,-8,-40,-48.

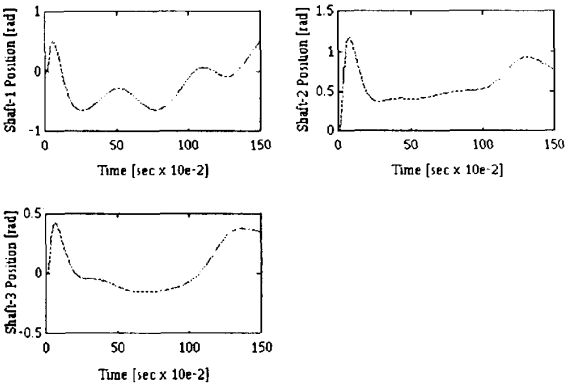


Fig.11-Shaft positions for the control in Fig.10.

V. Conclusion

In this work, feedback linearization as a control method for flexible joint robots is considered and its performance is compared to classical controllers.

Simulation results show that in the presence of joint flexibility, classical controllers fail to achieve tracking of reference trajectories. Even for set point control, the problem is significant. The feedback linearization method linearizes and decouples the dynamical equations of a flexible joint manipulator, so that any smooth trajectory is followed with accuracy.

The method includes a state space transformation which assigns the new state variables as link positions, velocities, accelerations and jerks. These variables are the ones to describe a time varying reference trajectory. Therefore a complicated inverse transformation is not necessary after doing the control for the new state variables.

A feedback linearization controller, even for simple configurations, requires many on line calculations to be performed to generate the control input from sensory information and the reference trajectory. This makes the use of fast computers necessary. Model simplification which can reduce the number of required calculations is an important advantage to be considered.

Control algorithms for the rigid robot are much simpler than the feedback linearization control. They require less computing power. If high accuracy is not a requirement, these methods can be used to control flexible robots in cases where the stiffness constant representing the joint flexibility is large.

The state variables of the new system obtained by feedback linearization are all used in feedback for the outer loop control. Position and velocity measurements are possible, but for link acceleration and jerk, the design of state observers is necessary.

The exact cancellation of nonlinearities can cause robustness problems. Robust outer loop control methods should be considered when using feedback linearization control.

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